



Oracle Collections & Arrays

PPT PRESENTED BY

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What are Oracle Collections?

Oracle Collections are single-dimensional, homogeneous arrays that store groups of elements. They are dynamic in-memory data structures utilized to batch and process bulk database rows.

Core Characteristics

- **Homogeneous:** All elements in a collection must have the exact same data type.
- **In-Memory:** Standard collection variables exist during PL/SQL execution, held in the session memory.
- **Dynamic Size:** Collections can shrink or grow at runtime using built-in API calls.
- **Bulk Operations:** Used to fetch multiple rows in a single step (BULK COLLECT) and bind arrays to SQL (FORALL).

Three Main Collection Types

- **Varrays (Variable-Size Arrays):** Have a fixed maximum size. Ordered lists starting at index 1. Can be stored in database table columns.
- **Nested Tables:** Unbounded size. Starts at index 1. Can become sparse (have gaps) when elements are deleted. Can be stored in the database.
- **Associative Arrays (Index-by Tables):** Unbounded size. Can be indexed by integers or strings (key-value maps). Exist in memory only. Cannot be stored in database tables.

Varrays (Variable-Size Arrays)

Varrays represent ordered, bounded lists of items. Ideal for small, fixed collections:

Syntax & Declaration

```
-- Define Varray type
TYPE phone_list IS VARRAY(3) OF VARCHAR2(15);

-- Declare and Initialize variable
my_phones phone_list := phone_list('123-456', '789-012');

-- Accessing element
first_phone := my_phones(1); -- 1-indexed
```

Key Rules & Limitations

- **Strict Boundary:** Cannot add more elements than the limit declared (e.g. VARRAY(3) cannot hold 4 elements).
- **Sequential Order:** Cannot delete individual elements from the middle to make it sparse. You can only delete from the end.
- **Constructor Initialization:** Must be initialized using a constructor before use. Using an uninitialized varray raises a COLLECTION_IS_NULL exception.
- **DB Storage:** Stored inline in database tables if small, preserving the exact element sequence.

Nested Tables vs. Associative Arrays

Understanding the core differences between unbounded collection types:

Feature	Nested Tables	Associative Arrays
Database Storage	Can be stored in DB table columns	PL/SQL memory only (cannot be in DB)
Indexing Key	Sequential integers starting at 1	Integers or VARCHAR2 strings (key-value)
Sparseness	Can become sparse (using DELETE(i))	Naturally sparse (keys can be non-sequential)
Initialization	Requires constructor: nested_tab_var()	No constructor; initialized on first assignment
Dynamic Growth	Requires EXTEND to allocate memory	Grows automatically as new keys are added

Oracle Collection API Methods

Collections are manipulated using built-in method functions:

Core Collection Methods Table

Method	Syntax Example	Description & Functionality
COUNT	var.COUNT	Returns the number of elements currently stored in the collection.
EXISTS	var.EXISTS(i)	Returns TRUE if element at index i is initialized; FALSE otherwise.
FIRST / LAST	var.FIRST	Returns the first (smallest) and last (largest) index keys in the collection.
EXTEND	var.EXTEND(n)	Allocates n new null elements at the end of Nested Tables / Varrays.
DELETE	var.DELETE(i)	Removes the element at index i, making the collection sparse.

Applications: Bulk SQL Processing

The primary use of collections is optimizing database queries using bulk operations:

BULK COLLECT Statement

Fetches an entire SQL query result set directly into a collection variable in a single context switch. Dramatically faster than fetching rows one-by-one in a cursor loop.

FORALL Statement

Sends an entire array of DML operations (INSERT, UPDATE, DELETE) to the SQL engine at once. Eliminates CPU context switches between PL/SQL and SQL engines.

Data Integration Caches

Used as temporary in-memory tables to join or aggregate data fetched from external web services before inserting into tables.

Stored Procedure Parameters

Enables client programs (Java, C#) to pass arrays of records to stored procedures, executing batch inserts in a single call.



THANK YOU

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